

Stability of approximate quantum broadcasting

Results and limits of a finite research campaign

ASTRA ULTRA campaign*

9 October 2026

Abstract

The campaign did not establish a historic-scale breakthrough. Its strongest internally checked results concern approximate quantum broadcasting. First, a deduction from Kitaev’s approximate factorization theorem gives a dimension-independent, linear bound on the distance of a self-compatible, nearly idempotent channel to an entanglement-breaking channel. An explicit antisymmetric example proves that a weaker, unentangled-input hypothesis cannot replace the complete norm. Second, reconstruction of an archived pure-state argument, a pairwise coherence estimate, and a second-order Gram-matrix measurement construction improve its reported fourth-root modulus to a linear modulus. An elementary lower example shows that the linear order cannot be improved uniformly. Complete deductions, assumptions, resource boundaries, and unresolved novelty questions are recorded below. Neither result establishes a new physical phenomenon or a dimension-independent computational capability.

1 Status and provenance

The mathematical scope is finite-dimensional complex quantum systems. A channel is completely positive and trace preserving (CPTP). State distance is

$$T(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1.$$

Channel distance below is the *full* diamond norm, without a factor of one half. An entanglement-breaking (EB) channel has a measure-and-prepare realization. “Internally checked” means a written derivation was reconstructed and challenged by another model investigator. It does not mean proof-assistant certification, external expert review, or verified historical priority.

Two immutable repositories were consulted selectively:

- Astra: 7a841813b40ffeb45addeba5969e323a71cbeab5.
- OpenAI Math: fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb.

Astra’s 00_START_HERE.txt was the initial entryptpoint. The pure-state argument reconstructs card N407 and its exact underlying proof (SHA-256 beginning 984acfe2c372). Its archived status was unreviewed; no repository assertion was promoted merely by retrieval. The channel factorization result below is imported from the specific primary preprint [1]; its 48-page proof was not independently rebuilt in full during this campaign. This dependency is explicit throughout.

The accompanying campaign ledger records the independent investigations, including failed and known-result routes. Computational receipts establish only their finite tested cases. The finite campaign created no Goal, automation, recurring run, or continuing background programme.

*AI-assisted mathematical research. Internal reconstruction and review are not independent external validation. No publication or priority claim is made.

2 Nearly idempotent broadcastable channels

A channel $\Phi : \mathcal{T}(H) \rightarrow \mathcal{T}(H)$ is self-compatible if one channel $\mathcal{C} : \mathcal{T}(H) \rightarrow \mathcal{T}(H \otimes H)$ has both marginals Φ . This requirement concerns the entire channel, including reference systems. It is substantially stronger than broadcasting a specified state family.

2.1 The precise imported result

Kitaev's Theorem 12.3 [1], printed page 44, supplies the following interface. There are universal constants C_1, C_2 and a positive universal smallness threshold such that, for a finite-dimensional CPTP Φ with

$$\eta = \|\Phi^2 - \Phi\|_\diamond$$

sufficiently small, there exist a finite-dimensional C^* -algebra A and channels

$$D : \mathcal{T}(H) \longrightarrow A_*, \quad E : A_* \longrightarrow \mathcal{T}(H)$$

with

$$\beta := \|\Phi - ED\|_\diamond \leq C_1\eta, \quad \gamma := \|DE - \text{id}_{A_*}\|_\diamond \leq C_2\eta. \quad (1)$$

Here A_* denotes the direct-sum state space of A . To check orientation, the source gives UCP maps $\Delta : A \rightarrow B(H)$ and $\Upsilon : B(H) \rightarrow A$ for Φ^* ; take $D = \Delta_*$ and $E = \Upsilon_*$. The source's equation (12.24), with one argument the unit, gives the second bound. These are complete norms. The constants are implicit in the source; this report does not provide numerical values.

The source audit found a local qualification in the displayed proof, on printed pages 27 and 45: combining arbitrary component unitary one-designs does not automatically give a diagonal for their direct sum. For $A = \mathbb{C} \oplus \mathbb{C}$, the singleton designs $\{1\}$ in each component give $1_A \otimes 1_A$, which fails the required centrality identity. Independently adjoining both signs to each component design preserves its own diagonal and makes its mean zero, eliminating the unwanted cross-block terms in the product ensemble. This repairs that construction with the same norm bounds. It is not a full revalidation of the source theorem; Theorem 2 remains explicitly conditional on its stated interface.

Lemma 1 (Logical no-cloning gap). *If a channel $\mathcal{Q} : \mathcal{T}(\mathbb{C}^m) \rightarrow \mathcal{T}(\mathbb{C}^m \otimes \mathbb{C}^m)$ has marginal diamond distances d_1, d_2 from the identity, then*

$$d_1 + d_2 \geq 2(1 - 1/m).$$

In particular, a qubit broadcaster must have at least one marginal distance at least $1/2$.

Proof. Apply $\text{id}_R \otimes \mathcal{Q}$ to the normalized maximally entangled state Ω_m . In the resulting state ω_{R12} , let $P_1 = \Omega_{R1} \otimes I_2$ and $P_2 = \Omega_{R2} \otimes I_1$. The diamond hypotheses imply $\text{Tr}(P_j \omega) \geq 1 - d_j/2$. The isometries $x \mapsto |\Omega_m\rangle_{R1} \otimes x_2$ and $y \mapsto |\Omega_m\rangle_{R2} \otimes y_1$ have cross-Gram operator I/m (after identifying the bases). Thus the sum of their range projections has norm $1 + 1/m$. Consequently

$$2 - \frac{1}{2}(d_1 + d_2) \leq \text{Tr}((P_1 + P_2)\omega) \leq 1 + 1/m.$$

Rearrange. This argument is a reconstruction of a standard no-cloning obstruction, not a claim of novelty for that obstruction. $\square$

Theorem 2 (A consequence of Kitaev factorization). *Under (1), suppose a joint channel has marginals Φ_1, Φ_2 satisfying $\|\Phi_j - \Phi\|_\diamond \leq \tau_j$. If*

$$\tau_1 + \tau_2 + 2\beta + 4\gamma < 1, \quad (2)$$

then A is commutative and $\Lambda = ED$ is entanglement breaking, with $\|\Phi - \Lambda\|_\diamond \leq \beta$. In particular, there are universal $C, \eta_0 > 0$ such that every self-compatible Φ with $\eta < \eta_0$ is within $C\eta$ of EB in diamond norm.

Proof. Suppose A has a summand M_m with $m \geq 2$. Embed a qubit into a two-dimensional corner of that summand by a CPTP map i . There is a CPTP retraction r with $ri = \text{id}$: compress the selected corner and send all remaining trace to a fixed qubit state. These operations are channels on the direct-sum state spaces, also when extended by a reference.

Let $\mathcal{C}$ denote the assumed joint channel. Then

$$\mathcal{Q} = (rD \otimes rD)\mathcal{C}Ei$$

is a physical qubit broadcaster. Its j th marginal is $rD\Phi_j Ei$. Since channels have diamond norm one,

$$\begin{aligned} \|D\Phi_j E - \text{id}\|_\diamond &\leq \tau_j + \|D(\Phi - ED)E\|_\diamond + \|(DE)^2 - \text{id}\|_\diamond \\ &\leq \tau_j + \beta + 2\gamma. \end{aligned}$$

The last bound follows by writing $(DE)^2 - \text{id} = DE(DE - \text{id}) + (DE - \text{id})$. Composition with i, r preserves this upper bound. Lemma 1 for $m = 2$ now contradicts (2).

Therefore all simple summands of A are one-dimensional: $A \cong \mathbb{C}^k$. The channel D is a measurement and E prepares a state from its classical outcome. Thus ED is EB and (1) gives the approximation. For exact self-compatibility set $\tau_1 = \tau_2 = 0$ and choose a universal threshold with $(2C_1 + 4C_2)\eta_0 < 1$, also inside the source theorem's range. $\square$

This is a complete deduction *from the stated imported theorem*. Its historical priority is unresolved. The approximating channel is not asserted to be idempotent; recent primary work reports an obstruction to that stronger conclusion [3].

2.2 The linear order cannot be improved

The known mirror phase-covariant cloner [7] supplies an elementary sharpness family. For $0 < e \leq 1/16$ and $|S\rangle = (|01\rangle + |10\rangle)/\sqrt{2}$, define an isometry

$$V|0\rangle = \sqrt{1-e}|00\rangle|0\rangle_E + \sqrt{e}|S\rangle|1\rangle_E, \quad V|1\rangle = \sqrt{1-e}|11\rangle|1\rangle_E + \sqrt{e}|S\rangle|0\rangle_E.$$

Its two output marginals agree. Their channel T_e has Pauli multipliers (s, s, z) , where $s = \sqrt{2e(1-e)}$ and $z = 1 - e$. Put $u = s - s^2$ and $v = e(1-e) = s^2/2$. The coefficients of $T_e^2 - T_e$ in the Pauli conjugation basis (I, X, Y, Z) are

$$\frac{1}{4}(- (2u + v), v, v, 2u - v).$$

For a real Pauli-conjugation expansion, the diamond norm equals the sum of the absolute coefficients: the triangle inequality proves the upper bound, and a Bell input attains it because its four Pauli images are orthogonal. Here $2u - v > 0$, and therefore

$$\eta_e = \|T_e^2 - T_e\|_\diamond = s - \frac{3}{4}s^2.$$

The EB channel $(1-e)\mathcal{D}_Z + (e/2)(\mathcal{D}_X + \mathcal{D}_Y)$, with $\mathcal{D}_P$ dephasing in the P basis, has distance $s - e/2$ from T_e by the same coefficient calculation. Conversely the Bell overlap of the normalized

Choi state of T_e is $1/2 + (s - e/2)/2$, while every separable Choi state has overlap at most $1/2$. This witness proves

$$\text{dist}_\diamond(T_e, \text{EB}) = s - e/2, \quad \frac{\text{dist}_\diamond(T_e, \text{EB})}{\eta_e} \longrightarrow 1.$$

Thus no uniform $o(\eta)$ modulus is possible, even for qubits. The cloner family is prior art; priority of this stability comparison has not been established.

2.3 Why unentangled tests cannot replace the diamond hypothesis

Proposition 3. *For each $d \geq 3$ there is a self-compatible channel W_d whose idempotence defect on every unentangled input tends to zero, while its diamond distance from every EB channel is at least one.*

Proof. Set

$$W_d(X) = \frac{\text{Tr}(X)I - X^\top}{d - 1}.$$

Its normalized Choi state is $(I - F)/[d(d - 1)]$, where F swaps the two factors. It is positive and has maximally mixed input marginal, hence defines a channel. The maximally mixed state on $\bigwedge^3 \mathbb{C}^d$ has this same state as each two-body marginal and I/d as its reference marginal. The Choi correspondence therefore supplies a joint channel with both marginals W_d .

Write $\mathcal{D}(X) = \text{Tr}(X)I/d$, $\mathsf{T}(X) = X^\top$, and $a = (d - 1)^{-1}$. Direct calculation gives

$$W_d = \mathcal{D} - a(\mathsf{T} - \mathcal{D}), \quad W_d^2 = \mathcal{D} + a^2(\text{id} - \mathcal{D}), \quad W_d^2 - W_d = a(\mathsf{T} - \mathcal{D}) + a^2(\text{id} - \mathcal{D}).$$

Thus every density matrix satisfies $\|(W_d^2 - W_d)(\rho)\|_1 \leq 2a + 2a^2 \rightarrow 0$.

The antisymmetric projector $P_- = (I - F)/2$ has probability one on the Choi state of W_d . On any product pure state its probability is $(1 - |\langle x, y \rangle|^2)/2 \leq 1/2$, so the same upper bound holds on every separable state. EB Choi states are separable. This single measurement gives normalized Choi trace distance at least one, and hence $\|W_d - \Lambda\|_\diamond \geq 1$ for every EB Λ .

There is no contradiction with Theorem 2. The Choi state of W_d^2 is $(1 - a^2)I/d^2 + a^2\Omega_d$; its antisymmetric probability is $(d - 2)/[2(d - 1)]$. Therefore the complete idempotence defect is at least $d/(d - 1)$, not a vanishing quantity. $\square$

2.4 Consequences and resource limits

For any reference system, replacing one use of Φ by Λ changes the joint state in trace norm by at most β . Replacing t uses costs at most $t\beta$ by telescoping. Also $\Lambda\Phi^{t-1}$ is EB and within β of Φ^t for every $t \geq 1$. These are mathematical channel comparisons, not claims about a measured physical apparatus.

When $A = \mathbb{C}^k$, the classical transition DE retains each label with probability at least $1 - \gamma/2$. The conclusion does not identify a physical preferred basis, prove independent fragments in the actual broadcaster, or solve tomography. The factorization is existential here: obtaining Φ , certifying a diamond defect, constructing D, E , implementing measurements, and establishing numerical source constants are distinct acquisition and computation tasks. Their costs are not dimension independent.

3 Uniform reconstruction of promised pure states

The second result starts with a weaker input promise and gives a correspondingly weaker output norm. It does not follow from Theorem 2: no global near-idempotence hypothesis is assumed.

Theorem 4 (Explicit pure-family stability). *Let H be finite-dimensional and $\mathcal{F}$ any family of pure states. Suppose one full-domain channel $\mathcal{C} : H \rightarrow H \otimes H$ has both marginal errors at most b on every member of $\mathcal{F}$. For $0 \leq b \leq 1/4096$, define*

$$\delta = 4\sqrt{b}, \quad \eta = \frac{3}{2}((1 - 4\delta)^{-1/2} - 1), \quad a = \delta + \eta.$$

There is one EB channel with uniform half-trace error at most

$$\kappa(b) = (\sqrt{a\eta} + \sqrt{2b})^2, \quad f(b) = \kappa(b) + (1 - \kappa(b))14b. \quad (3)$$

For larger b use the bound one. In particular $f(b) = (208 + 16\sqrt{6})b + O(b^{3/2})$ as $b \downarrow 0$. The bound is uniform in dimension, family cardinality and prior; it is not a diamond approximation to the broadcaster. The linear order is optimal, without a claim that the displayed constant is optimal.

3.1 A reconstructed Schur partition

Fix a finite subfamily $P_i = |\psi_i\rangle\langle\psi_i|$ and a Stinespring isometry V for the broadcaster. Let $u_i = V\psi_i$ and $K_i = P_i \otimes P_i \otimes I$. Commutativity of the two local target tests gives $q_i = \langle u_i, K_i u_i \rangle \geq 1 - 2b$. Normalize the projection to write $v_i = \psi_i \otimes \psi_i \otimes e_i$ with $\langle v_i, u_i \rangle = \sqrt{q_i}$. Then $\|u_i - v_i\| \leq 2\sqrt{b}$.

Put $C_{ij} = \langle \psi_i, \psi_j \rangle$, $F_{ij} = \langle e_i, e_j \rangle$ and $L = C \circ F$. Their Schur multipliers are UCP. Diagonal isometries $W_u e_i = e_i \otimes u_i$ and $W_v e_i = e_i \otimes v_i$ yield, by telescoping at every amplification,

$$\|S_C - S_{C \circ C \circ F}\|_{\text{cb}} \leq \delta.$$

Composition with S_F gives $\|S_L - S_L^2\|_{\text{cb}} \leq \delta$.

In the Banach algebra of Schur multipliers, set $Z = 2S_L - I$ and $R = (I + 4(S_L^2 - S_L))^{-1/2}$ using the convergent binomial series. Then $Q = (I + ZR)/2$ is idempotent and

$$\|Q - S_L\|_{\text{cb}} \leq \eta.$$

Its symbol is Hermitian with entries zero or one and diagonal one. Since $1 + \eta \leq \sqrt{3} - 1/2 < 5/4$, its ones form an equivalence relation. Indeed a failed transitivity would give the principal mask

$$M = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Applying its Schur map to the all-ones matrix divided by three gives trace norm $(1 + 2\sqrt{2})/3 > 5/4$, contradicting the multiplier norm bound (use the conjugate-symbol predual). Thus Q is a block pinching, and

$$\|S_C - S_{C \circ Q}\|_{\text{cb}} \leq a. \quad (4)$$

Within a block $|C_{ij}| \geq 1 - \eta > 3/4$. Across blocks, $|C_{ij}| \leq \delta/(1 - \eta) < 0.082$. Therefore the partition is exactly the relation $\text{Tr}(P_i P_j) > 1/4$.

3.2 The physical pair estimate

Lemma 5. *For any pair satisfying the broadcasting promise above, if $r = |\langle \psi, \phi \rangle| > 3/4$ and $b \leq 1/4096$, then $s = T(P_\psi, P_\phi) = \sqrt{1 - r^2} \leq 14b$.*

Proof. First decompose $u_i = \sqrt{q_i}v_i + w_i$ with $K_i w_i = 0$ and $\|w_i\| \leq \sqrt{2b}$. Orthogonality to the full target-output subspace implies

$$r \leq r^2 + 2\sqrt{2b}\sqrt{1 - r^2} + 2b.$$

Writing $s = \sqrt{1 - r^2}$ yields

$$\frac{r}{1+r} s^2 \leq 2\sqrt{2b} s \sqrt{1+r^2} + 2b \leq 4\sqrt{b} s + 2b.$$

Since $r/(1+r) > 3/7$, for $b > 0$ the variable $x = s/\sqrt{b}$ obeys $(3/7)x^2 \leq 4x + 2$, whence $x < 10$. Indeed $x \geq 10$ would give $(3/7)x \leq 4 + 2/x \leq 4.2$, impossible. Thus $s < 10/64 = 5/32 < 1/6$. The $b = 0$ case follows directly from the same inequality.

Choose a basis with $\psi = |0\rangle$, $\phi = r|0\rangle + s|1\rangle$, and write their joint outputs as ρ_0, ρ_1 . Each marginal off-diagonal entry differs from its target by at most b . Therefore the difference of the two output marginal coherences differs from rs by at most $2b$.

In passing from an output marginal coherence to the joint matrix entry between $|00\rangle$ and $|10\rangle$, its omitted sum is bounded, by Cauchy–Schwarz over the entire complementary second-system subspace, by the probability that the second system is outside $|0\rangle$. Those probabilities are at most b and $s^2 + b$ respectively. Hence each of

$$\langle 00 | (\rho_1 - \rho_0) | 10 \rangle, \quad \langle 00 | (\rho_1 - \rho_0) | 01 \rangle$$

has real part at least $rs - s^2 - 4b$. Test the joint output against the normalized symmetric coherence between $|00\rangle$ and $(|10\rangle + |01\rangle)/\sqrt{2}$. Trace-distance contraction gives

$$s \geq \sqrt{2}(rs - s^2 - 4b), \quad \text{or} \quad 4b \geq s(r - s - 1/\sqrt{2}). \quad (*)$$

This argument includes arbitrary output dimensions: no sum over discarded levels is bounded by their number.

From $s < 1/6$ the bracket in $(*)$ is greater than $1/9$, so $s \leq 36b < 1/100$. The bracket is then greater than $4/15$, giving $s \leq 15b < 1/250$. It is now greater than $2/7$, giving $s \leq 14b$. These inequalities follow by substituting $r = \sqrt{1 - s^2}$; each is a strict rational comparison after squaring positive quantities. $\square$

3.3 The second-order measurement construction

The ideal two-copy vectors v_i have Gram matrix $J = C \circ L$. Because Q is a zero-one block mask, the following identity holds entrywise:

$$J - J \circ Q = (C - C \circ Q) \circ (L - Q).$$

Schur multiplication is composition of Schur maps. Consequently

$$\|S_{J-J \circ Q}\|_{\text{cb}} \leq a\eta =: h. \quad (5)$$

This second-order estimate is the gain over measuring the input Gram matrix directly.

For any prior p on the finite subfamily, set $G = D_p J D_p$, where $D_p = \text{diag}(\sqrt{p_i})$, and $G_0 = G \circ Q$. These are density matrices. Equation (5) gives $\|G - G_0\|_1 \leq h$: apply the trace-class Schur map to zz^* with $z_i = \sqrt{p_i}$. Its dual symbol is the transpose, or complex conjugate, of $J - J \circ Q$; conjugation preserves its completely bounded norm. Thus the estimate applies also for complex Gram entries. The Powers–Størmer inequality gives $\|\sqrt{G} - \sqrt{G_0}\|_2^2 \leq h$. For $p_i > 0$, define $\alpha_i = \sqrt{G}e_i/\sqrt{p_i}$ and $\alpha_i^0 = \sqrt{G_0}e_i/\sqrt{p_i}$. The first vectors have Gram matrix J , and the second lie in the coordinate block of i . The map $v_i \mapsto \alpha_i$ therefore extends to an isometry on their span. Pulling back the coordinate-block projections gives a POVM on that span with prior-averaged wrong-block probability at most h . Extend it to all of $H \otimes H \otimes E$ by assigning the orthogonal complement to one fixed outcome. Zero-prior entries can be omitted.

For completeness, the trace square-root inequality used here has a short finite-dimensional proof. Put $A = \sqrt{G}$, $B = \sqrt{G_0}$, $X = A - B$, and $S = \text{sign}(X)$. Trace duality and $A^2 - B^2 = AX + XB$ imply $\|A^2 - B^2\|_1 \geq \text{Tr}|X|(A + B)$. In an eigenbasis of X , positivity of A, B gives $\langle v, (A + B)v \rangle \geq |\langle v, (A - B)v \rangle|$ for each eigenvector v ; therefore the last trace is at least $\text{Tr} X^2$.

Let N_c denote this output-space POVM and $R_i = I - N_{c(i)}$ its wrong-block effect on label i . Put $\epsilon_i = \langle v_i, R_i v_i \rangle$, so $\sum_i p_i \epsilon_i \leq h$. The actual Stinespring vectors obey $u_i = \sqrt{q_i} v_i + w_i$, with $\|w_i\|^2 \leq 2b$. Taking the square root of the effect, which need not be a projection, gives

$$\sqrt{\langle u_i, R_i u_i \rangle} = \|\sqrt{R_i} u_i\| \leq \sqrt{\epsilon_i} + \sqrt{2b}.$$

Average the squares and use Cauchy–Schwarz to obtain

$$\sum_i p_i \langle u_i, R_i u_i \rangle \leq (\sqrt{h} + \sqrt{2b})^2 = \kappa(b).$$

The operators $M_c = V^* N_c V$ form a POVM on the original input space and have precisely these error probabilities. This existence proof requires no physical access to the Stinespring environment.

The partition Q is fixed independently of p . Wrong-block probability is bilinear in the prior and the input POVM. Minimax over the compact convex POVM set thus produces *one* input POVM with every wrong-block probability at most $\kappa(b)$. Fix a representative in each block and reprepare it on that block’s outcome. Lemma 5 bounds its correct-outcome error by $14b$, while a wrong outcome has error at most one. Convexity yields $\kappa(b) + (1 - \kappa(b))14b$. In the stated range, $\eta < 1/4$, $a < 1/3$ and $2b \leq 1/2048$ ensure $\kappa(b) < 1$.

For an arbitrary family in dimension d , the overlap relation is an equivalence relation because every finite triple has the proved partition. In fact there are at most $\lfloor d/(1 - \kappa(b)) \rfloor$ classes. To see this, choose representatives from any k distinct classes. Their input POVM satisfies $\text{Tr} M_c \geq \text{Tr}(M_c P_c) \geq 1 - \kappa(b)$; summing gives $k(1 - \kappa(b)) \leq d$. Thus only finitely many classes exist. Fix their representatives. For each finite subfamily together with these representatives, the preceding argument supplies a POVM with these same class labels. The set of k -outcome POVMs is compact, and each input imposes a closed wrong-class constraint. The finite-intersection property supplies one common POVM for the entire family. Repreparing the fixed representatives proves Theorem 4. The empty family is immediate.

3.4 Why the linear order is optimal

Let $0 < t \leq 1/2$, $\psi = |0\rangle$, $\phi = t|0\rangle + \sqrt{1 - t^2}|1\rangle$, and write their projectors as P, Q . Let b_t be the optimal uniform broadcasting error for this pair and e_t its optimal EB reconstruction error. Then

$$t^2/72 \leq b_t \leq e_t \leq t^2. \quad (6)$$

For the upper bound, measure in the computational basis and prepare $P \otimes P$ or $Q \otimes Q$. The marginals fix P and send Q to $(1 - t^2)Q + t^2P$, at error at most t^2 . The marginal channel is EB. Every EB channel can be broadcast by preparing two copies of its conditional output, so $b_t \leq e_t$.

For the lower bound, a broadcaster of error b has Stinespring vectors $u_\psi = \psi\psi \otimes z_\psi + r_\psi$ and $u_\phi = \phi\phi \otimes z_\phi + r_\phi$, with $\|z_i\| \leq 1$ and $\|r_i\| \leq \sqrt{2b}$ by the two target tests. Preservation of the input overlap implies

$$t \leq t^2 + 2\sqrt{2b} + 2b.$$

For $b \leq 1/2$, the last two terms are at most $3\sqrt{2b}$, giving $b \geq (t - t^2)^2/18 \geq t^2/72$; for $b > 1/2$ this lower bound is automatic. Taking the infimum proves (6). The feasible promises $b = t^2$ tend to zero while $e_t \geq b/72$, ruling out a uniform $o(b)$ bound. This is an elementary rate witness, not a novelty claim for two-state cloning [6].

3.5 The finite-list algorithm and its costs

If $P_1, \dots, P_n$ are explicitly supplied rational pure density matrices and the broadcasting promise is granted, compute the overlap partition and choose its $k \leq n$ representatives. Solve

$$\begin{aligned} & \text{minimize } t \\ & \text{subject to } M_c \succeq 0, \quad \sum_{c=1}^k M_c = I, \quad 0 \leq t \leq 1, \\ & \quad \text{Tr}(M_{c(i)} P_i) \geq 1 - t \quad (i = 1, \dots, n). \end{aligned}$$

The optimum is at most $\kappa(b)$. The resulting channel is $\Lambda(X) = \sum_c \text{Tr}(M_c X) P_{r(c)}$. This is a construction given a supplied state list and promise, not an algorithm for discovering that the promise holds. Standard broadcasting SDPs predate this campaign [4]; the new part under examination is the quantitative implication, not semidefinite optimization itself.

Naive overlap formation uses $O(n^2 d^2)$ rational scalar operations. The SDP has $O(kd^2)$ real variables, n input inequalities and k PSD blocks of order d . It has a strict feasible point $M_c = I/k$, $t = 1 - 1/(2k)$, with margin $1/(2k)$. Eliminating the equality constraint gives polynomially bounded radius and inverse interior radius; standard rational weak-SDP algorithms therefore support additive optimization tolerance in polynomial bit time in the explicit input size and logarithm of inverse tolerance. Exact optimal real matrices need not have finite rational representations. The included floating solver is a diagnostic implementation, not a certified rational solver.

An implemented rational POVM with additive wrong-label loss ϵ adds at most ϵ to the stated error. Unknown states require acquisition: tomography, state preparation, measurement synthesis, device calibration, and sample costs are not supplied by this theorem. No claim of dimension-free memory, time, energy, or experiment count is made. The arbitrary-family compactness result has no finite-list input representation.

4 What the campaign established and did not establish

The two results have different premises. Theorem 2 uses a complete near-idempotence condition and gives a complete channel approximation, through a substantial imported theorem. Theorem 4 uses only a promised pure-state family and gives a uniform statewise reconstruction with an explicit small-error modulus. Proposition 3 proves that a tempting weakening of the first premise fails. None of these statements settles general mixed-family approximate broadcasting, the best constant in the pure-family modulus, or a major open conjecture.

The pure-family pipeline was frozen, independently reconstructed, and then separately reviewed for its finite-list algorithm, the pairwise strengthening, and the second-order measurement construction. Finite quantum SDP cases include an explicit dephasing broadcaster, two-state qubit broadcasting and EB optimizations, and an unpromised nontransitive overlap chain. Some solves are labelled `optimal_inaccurate`; numerical tolerances and residuals remain in the receipts. These checks are falsification diagnostics, not universal proof or benchmark superiority.

Broader investigations produced source critiques, exact observational-equivalence examples, and conditional interventions across arithmetic, computation, inference, transport, chemistry, diffraction and spectroscopy. Their known predecessors, acquisition assumptions and failures are preserved in the campaign ledger. The arithmetic $H_{10}(\mathbb{Q})$ source remains unverified; a conditional dependency simplification is not a proof of that conjecture. No new empirical phenomenon or transformative operational capability was demonstrated. The mission’s historic-breakthrough criterion remains unmet.

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